

LiCl-Mediated Preparation of Highly
Functionalized Benzylic Zinc Chlorides

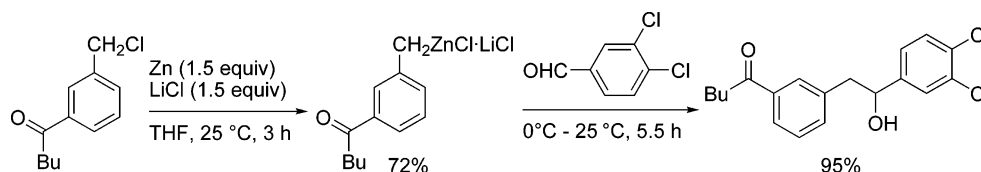
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ABSTRACT



In the presence of zinc dust (1.5–2.0 equiv) and LiCl (1.5–2.0 equiv), various benzylic chlorides bearing functional groups (iodide, cyanide, ester, ketone) are smoothly converted at 25 °C to the corresponding zinc reagents without homo-coupling (<5%). The utility of these benzylic zinc reagents is demonstrated by a short synthesis of papaverine.

Zinc organometallics are important intermediates since they combine an excellent functional group compatibility with a high reactivity (in the presence of the appropriate catalyst).¹ Especially functionalized benzylic zinc halides occupy a unique place since the high reactivity of related benzylic lithium and magnesium compounds preclude the presence of most functional groups in these organometallics. Benzylic zinc reagents can be prepared by the direct zinc insertion to benzylic bromides but may require elevated temperatures and the addition of polar cosolvents.² These drawbacks hamper a more general application of these organometallics. Recently, we have reported that LiCl facilitates considerably

the rate of zinc insertion.³ Herein, we report an efficient preparation of functionalized benzylic zinc chlorides of type **1** using the LiCl-mediated insertion of commercial zinc dust to functionalized benzylic chlorides of type **2** (Scheme 1 and

Scheme 1. LiCl-Mediated Insertion into Functionalized Benzylic Chlorides

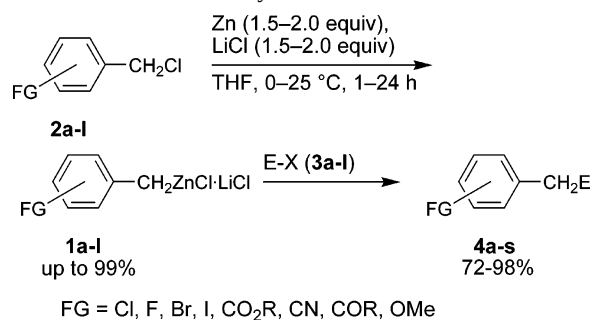


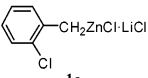
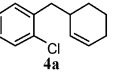
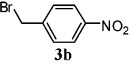
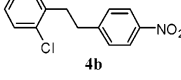
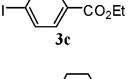
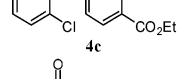
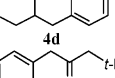
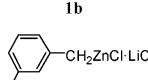
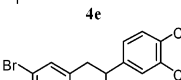
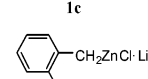
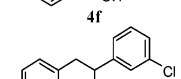
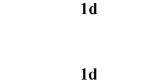
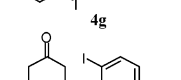
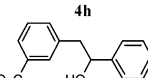
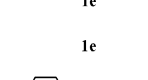
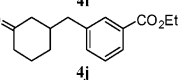
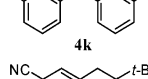
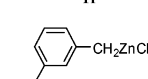
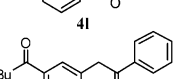
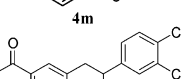
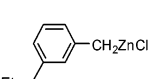
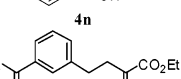
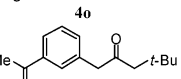
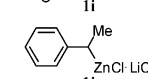
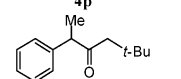
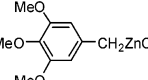
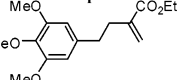
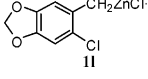
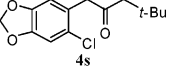
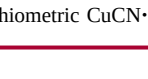
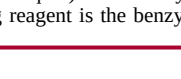
Table 1). Their reaction with various electrophiles (**3**) provides a range of polyfunctional products of type **4**. Thus, the addition of 2-chlorobenzyl chloride (**2a**, 1.0 equiv) to

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Table 1. LiCl-Mediated Insertion into Functionalized Benzylic Chlorides and Reaction with Various Electrophiles

entry	benzylic zinc chloride (1)	insertion (h) ^a , yield (%) ^b	electrophile (3)	temperature (°C), time (h)	product (4)	yield (%) ^d
1	 1a	2, 99	 3a	25, 1.5	 4a	94 ^{e,h}
2	1a		 3b	0, 3	 4b	89 ^c
3	1a		 3c	60 °C, 5	 4c	97 [#]
4	1a		 3d	−40 to 25, overnight	 4d	93 ^f
5	 1b	24 ^c , 87	 3e	−40 to 25, overnight	 4e	95 ^f
6	 1c	4, 95	 3f	0 to 25, 17	 4f	98
7	 1d	2, 99	 3g	0 to 25, 5	 4g	87
8	1d		 3d	−40 to 25, overnight	 4h	72 ^f
9	 1e	3.5 ^c , 85	 3h	0 to 25, 4.5	 4i	91
10	1e		 3d	−40 to 25, overnight	 4j	97 ^f
11	 1f	3, 93	 3i	60, 5	 4k	88 [#]
12	1f		 3j	−60 to −20, overnight	 4l	78 ^f
13	 1g	3.5, 72	 3k	−20, overnight	 4m	85 ^f
14	1g		 3f	0 to 25, 5.5	 4n	95
15	 1h	3, 72	 3l	−60 to 0, 1	 4o	92 ^e
16	 1i	3.5, 68	 3e	−60 to −20, overnight	 4p	74 ^f
17	 1j	11, 85	 3e	−60 to 25, overnight	 4q	96 ^f
18	 1k	3 ^c , 78	 3l	−40 to 0, 1	 4r	98 ^c
19	 1l	1, 93	 3e	−60 to 25, overnight	 4s	93 ^f

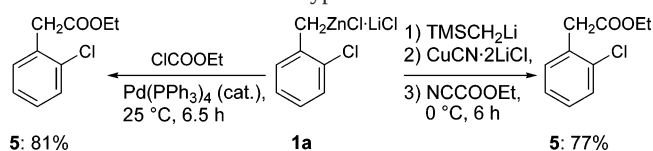
^a Reaction time at 25 °C. ^b Yield as determined by iodometric titration. ^c Zn (2.0 equiv), LiCl (2.0 equiv). ^d Isolated yield of analytically pure product. ^e Catalytic CuCN·2LiCl. ^f Stoichiometric CuCN·2LiCl. ^g Pd(PPh₃)₄ (cat.). ^h In this case, the limiting reagent is the benzylic zinc chloride **1a**.

zinc dust⁴ (1.5 equiv) and LiCl (1.5 equiv) at 0 °C followed by 2 h of stirring at 25 °C provides almost quantitatively 2-chlorobenzylzinc chloride **1a** (in 99% yield as determined by iodometric titration).⁵ In the absence of LiCl, only a slow and incomplete reaction is observed under these reaction conditions. The benzylic zinc reagent **1a** undergoes a range of useful reactions with electrophiles. Thus, Cu(I)-catalyzed cross-couplings with allylic and benzylic bromide such as 3-bromo-1-cyclohexene (**3a**, 1.3 equiv) or 4-nitrobenzyl bromide (**3b**, 0.8 equiv) afford the cross-coupling products **4a** and **4b** in 89–94% yield (entries 1 and 2 of Table 1).

Also, Negishi⁶ cross-coupling of ethyl 4-iodobenzoate (**3c**, 0.8 equiv) in the presence of Pd(PPh₃)₄ (2 mol %, 60 °C, 5 h) gives the diarylmethane derivative **4c** in 97% yield (entry 3). A copper(I)-mediated 1,4-addition of cyclohexenone (0.8 equiv) using CuCN·2LiCl⁴ (1 equiv, −40 °C to 25 °C, overnight) and TMSCl⁷ (2 equiv) provides the Michael adduct **4d** in 93% yield (entry 4). Similarly the 4-fluoro-, 3-bromo-, and 2-iodobenzyl zinc chlorides (**1b–d**) were prepared in 87–99% yield. The Cu(I)-mediated reaction of the benzyl zinc chloride **1b** with 3,3-dimethylbutyryl chloride (**3e**, 0.7 equiv) in the presence of CuCN·2LiCl (1.0 equiv, −40 °C to 25 °C, overnight) leads to the ketone **4e** in 95% yield. The high reactivity of benzyl zinc chlorides allows an efficient addition to various aldehydes *in the absence of any catalyst*. Thus, the addition of the substituted benzaldehydes (**3f** or **3g**; 0.8 equiv, 0–25 °C, 5–17 h) gave the benzyl alcohols **4f** and **4g** in 87–98% yield (entries 6 and 7). The Cu(I)-mediated Michael addition of **1d** to cyclohexenone furnishes an expected ketone **4h** in 72% yield (entry 8). Various benzyl zinc reagents bearing carbonyl functions have also been prepared (entries 9–16). Thus, the reaction of substituted benzyl chlorides bearing a carbethoxy group, a cyanide, and a ketone group in the meta-position with zinc dust at 25 °C provides smoothly the corresponding zinc reagents (**1e–i**; entries 9–16). Remarkably, the keto group present in the benzyl zinc chlorides **1g–i** is quite stable toward enolization. Thus, the 3-propionyl benzyl zinc chloride (**1h**) has a half-life of 27 days at 25 °C. Even the acetyl-substituted benzyl zinc reagent **1i** is stable for several days ($t_{1/2}$ = 2 days, 25 °C). These zinc reagents react smoothly with typical electrophiles, leading to the functionalized products **4i–p** (74–97%; entries 9–16). Secondary benzyl zinc compounds can be prepared similarly. Thus, the treatment of 1-chloroethylbenzene with zinc dust (1.5 equiv) and LiCl (1.5 equiv) at 25 °C for 11 h provides the expected zinc reagent **1j** in 85% yield. Its acylation with the acid chloride **3e** provides the ketone **4q** in 96% yield. In contrast, cumyl chloride (a tertiary benzyl chloride) did not afford the corresponding zinc species due to competitive elimination. Electron-rich benzyl chlorides,

such as 3,4,5-trimethoxybenzyl chloride are easily converted within 3 h at 25 °C (zinc dust 2.0 equiv, LiCl 2.0 equiv) to the benzylic zinc compound (**1k**, 78%). The allylation of **1k** with ethyl (2-bromomethyl)acrylate⁸ (0.8 equiv) affords the desired product **4r** in 98% yield (entry 18). Also, the electron-rich benzylic chloride **2i** is readily converted to the corresponding zinc compound **1l** within 1 h (zinc dust 1.5 equiv, LiCl 1.5 equiv) in 93% yield. Acylation of **1l** leads to the ketone **4s** in 93% yield (entry 19). Phenylacetic acid derivatives are common targets in pharmaceutical research.⁹ They are readily prepared in two ways using benzylic zinc species of type **1** (Scheme 2). Thus, the Negishi Pd-catalyzed

Scheme 2. Synthesis of Phenylacetic Acid Derivatives of Type 5

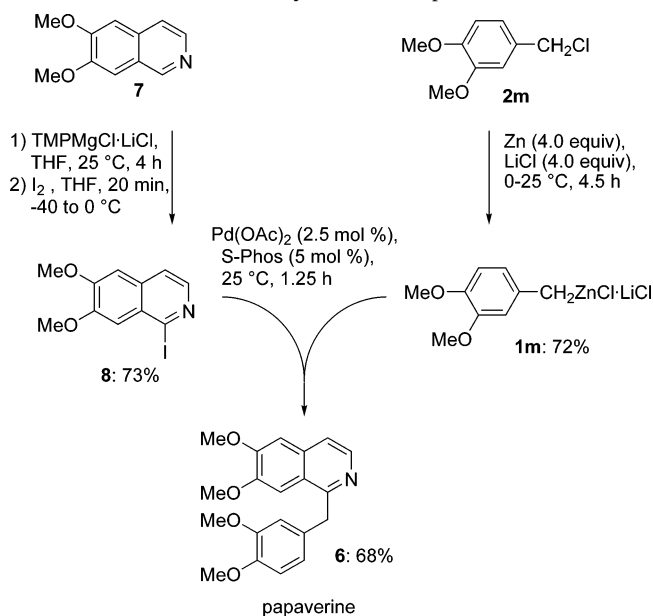


acylation¹⁰ of the benzylic zinc chloride **1a** with ethyl chloroformate in the presence of Pd(PPh₃)₄ (5 mol %, 25 °C, 6.5 h) is leading to the phenylacetic acid ethyl ester **5** in 81% yield. Alternatively, a copper mediated acylation is possible.

Thus, the reaction of **1a** with TMSCH₂Li (−30 °C, 30 min) provides a mixed diorganozinc of the type ArCH₂·ZnCH₂SiMe₃.¹¹ Transmetalation to copper (CuCN·2LiCl, −30 °C, 30 min) and addition of Mander's reagent¹² gives the expected ethyl phenylacetic ester **5** in 77% yield.

As an application, we have prepared papaverine (**6**)¹³ starting from the isoquinoline¹⁴ **7** (Scheme 3). Magnesiation

Scheme 3. Synthesis of Papaverine



(4) See Supporting Information, see also: Knochel, P.; Yeh, M. C. P.; Berk, S. C.; Talbert, J. J. *Org. Chem.* **1988**, 53, 2390.

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of **7** with $\text{TMPMgCl} \cdot \text{LiCl}$ ¹⁵ (TMP = 2,2,6,6-tetramethylpiperidyl; 25 °C, 4 h), followed by iodolysis provides the isoquinoline **8** in 73% yield. Zinc insertion to the benzylic chloride **2m** (25 °C, 4.5 h) gives the expected benzylic zinc reagent **1m** in 72% yield. Pd-catalyzed cross-coupling using $\text{Pd}(\text{OAc})_2$ (2.5 mol %) and S-Phos¹⁶ (5 mol %) provides papaverine (**6**) in 68% yield.¹⁷

In summary, we have reported a mild method allowing the preparation of benzylic zinc chlorides from the corresponding benzylic chlorides using zinc dust and LiCl. Remarkably, these zinc organometallics are compatible with

the presence of functional groups including ketones. Further extension of this work to the preparation of heterocyclic analogs is underway.

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Supporting Information Available: Experimental procedures and full characterization of all compounds. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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(11) ArCH_2ZnCl is not reactive enough and the mixed reagent $\text{ArCH}_2\text{-ZnCH}_2\text{SiMe}_3$ gives better results; see also: Berger, S.; Langer, F.; Lutz, C.; Knochel, P.; Mobley, T. A.; Reddy, C. K. *Angew. Chem., Int. Ed.* **1997**, 36, 1496.

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(17) The copper-catalyzed cross-coupling with the magnesiated isoquinoline (**7**) and 3,4-dimethoxybenzyl chloride (**2m**) did not provide the expected papaverine (**6**).